

Spatial Context Networks: Geometric Semantic Routing in Neural Architectures

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Abstract

We introduce *Spatial Context Networks* (SCN), a novel neural architecture that treats neurons as geometric entities in a learned semantic space. Unlike traditional neural networks that rely on weighted summations, SCN employs distance-based activation functions where each neuron operates as a point-mass with a learnable centroid in d -dimensional space. The architecture implements three key innovations: (1) geometric activation functions based on Euclidean distance, (2) semantic routing that selectively activates neurons based on spatial proximity, and (3) connection density weighting with adaptive scaling. Our experiments demonstrate stable training dynamics, interpretable neuron specialization, and efficient sparse activation patterns. Notably, all experiments were conducted on consumer-grade hardware (gaming laptop), demonstrating the accessibility and computational efficiency of this approach. The architecture achieves 91% network efficiency with only 32 hidden neurons while maintaining numerical stability through principled geometric constraints.

Keywords: Geometric Deep Learning, Sparse Neural Networks, Semantic Routing, Distance-Based Activations, Efficient Architectures

1 Introduction

Traditional neural networks operate through weighted linear combinations followed by nonlinear activation functions. This paradigm, while powerful, treats neurons as abstract computational units without explicit geometric or semantic interpretation. Recent advances in geometric deep learning [1] have shown promise in incorporating spatial and structural inductive biases, but primarily focus on graph-structured data rather than the internal organization of neural representations.

We propose a fundamentally different approach: treating each neuron as a *geometric sensor* in a learned seman-

tic space. This perspective offers several advantages:

1. **Interpretability:** Neurons have explicit positions (centroids) in semantic space, making their specialization geometrically interpretable.
2. **Sparsity:** Distance-based routing naturally induces sparse activation patterns without explicit regularization.
3. **Stability:** Geometric constraints provide natural bounds on activation magnitudes.
4. **Efficiency:** Selective routing reduces computational cost by activating only semantically relevant neurons.

Our contributions are:

- A novel geometric activation function based on normalized Euclidean distance
- A semantic routing mechanism for conditional neuron activation
- A connection density framework that adaptively scales based on network sparsity
- Empirical analysis demonstrating stable dynamics and efficient computation on consumer hardware

2 Related Work

2.1 Sparse Neural Networks

Sparse activation has been explored through various mechanisms including dropout [2], sparse coding [3], and mixture of experts [4]. Our approach differs by deriving sparsity from geometric principles rather than stochastic or learned gating mechanisms.

2.2 Geometric Deep Learning

Graph neural networks [5] and geometric deep learning frameworks [1] exploit geometric structure in input data. We extend this by imposing geometric structure on the *internal representations* of the network itself.

2.3 Distance-Based Methods

Radial basis function (RBF) networks [6] use distance-based activations but typically employ fixed centers. Our learnable centroids in high-dimensional semantic space provide greater flexibility and expressiveness.

2.4 Routing Mechanisms

Capsule networks [7] and mixture of experts [8] employ routing mechanisms. Our semantic routing operates through geometric distance in learned representation space rather than agreement or learned gates.

3 Methodology

3.1 Geometric Activation Function

We define the activation of a neuron as inversely proportional to the normalized distance from its centroid:

$$f(\mathbf{v}) = \frac{1}{\frac{\|\mathbf{v} - \boldsymbol{\mu}\|_2}{\sqrt{d}} + \epsilon} \quad (1)$$

where:

- $\mathbf{v} \in \mathbb{R}^d$ is the input vector
- $\boldsymbol{\mu} \in \mathbb{R}^d$ is the learnable centroid
- d is the dimensionality of the semantic space
- $\epsilon = \frac{1}{\text{SF}}$ is a stability factor (SF = 10 in our experiments)

The normalization by $\sqrt{d}$ ensures scale invariance across different dimensionalities, preventing the curse of dimensionality from dominating the distance metric.

Key Properties:

1. **Bounded:** $f(\mathbf{v}) \in (0, \frac{1}{\epsilon}]$ with maximum at $\mathbf{v} = \boldsymbol{\mu}$
2. **Smooth:** Differentiable everywhere except at the centroid (regularized by ϵ)
3. **Localized:** Activation decreases monotonically with distance

3.2 Semantic Routing Layer

The semantic routing layer selectively activates neurons based on their geometric affinity to the input:

$$S = \{n_i \mid f_i(\mathbf{q}) > \tau\} \quad (2)$$

where τ is the routing threshold (set to 0.5 in our experiments) and f_i is the geometric activation function for neuron i .

The active neuron mask is computed as:

$$M_{ij} = \mathbb{I}[f_j(\mathbf{v}_i) > \tau] \quad (3)$$

where $\mathbb{I}[\cdot]$ is the indicator function.

This binary masking creates discrete routing decisions, inducing structured sparsity in the network. Unlike soft attention mechanisms, hard routing reduces computational cost during inference.

3.3 Connection Density Weighting

To prevent activation collapse and incorporate historical context, we introduce a connection density term:

$$C = \frac{\sum_{i \in S} w_i}{\alpha/z} \quad (4)$$

where:

- w_i are learnable connection weights
- α is the total number of neurons (32 in our experiments)
- $z = |S|$ is the number of active neurons

The denominator α/z provides adaptive normalization: when fewer neurons are active (smaller z), the denominator increases, naturally dampening the signal. This serves as an automatic stabilization mechanism.

Explosion Control: To prevent unbounded growth, we apply a square root damping when the context score exceeds a threshold:

$$C_{\text{stable}} = \begin{cases} C & \text{if } C < \tau_{\text{exp}} \\ \sqrt{C} & \text{otherwise} \end{cases} \quad (5)$$

where $\tau_{\text{exp}} = 2.0$ in our experiments.

3.4 Probabilistic Pattern Distribution

The final output incorporates Bayesian priors through element-wise multiplication:

$$\mathbf{o} = \mathbf{a} \odot \text{softmax}(\mathbf{w}_p) \quad (6)$$

where $\mathbf{w}_p \in \mathbb{R}^{d_{\text{out}}}$ are learnable pattern weights and $\odot$ denotes the Hadamard (element-wise) product.

This formulation allows the network to learn which output patterns are a priori more likely, providing a probabilistic interpretation of the output space.

3.5 Complete Architecture

The full forward pass is:

$$\mathbf{a}_{\text{routed}}, M = \text{SemanticRouting}(\mathbf{x}) \quad (7)$$

$$c = \text{ConnectionDensity}(\mathbf{a}_{\text{routed}}, M) \quad (8)$$

$$\mathbf{h} = \text{Linear}(c) \quad (9)$$

$$\mathbf{o} = \mathbf{h} \odot \text{softmax}(\mathbf{w}_p) \quad (10)$$

Algorithm 1 provides the complete pseudocode.

Algorithm 1 Spatial Context Network Forward Pass

Require: Input $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$, centroids $\{\mu_i\}_{i=1}^n$, threshold τ

Ensure: Output $\mathbf{o} \in \mathbb{R}^{d_{\text{out}}}$

```

1: for  $i = 1$  to  $n$  do
2:    $a_i \leftarrow \frac{1}{\|\mathbf{x} - \mu_i\|_2 + \epsilon}$ 
3:    $m_i \leftarrow \mathbb{I}[a_i > \tau]$ 
4: end for
5:  $z \leftarrow \sum_{i=1}^n m_i$ 
6:  $c \leftarrow \frac{\sum_{i=1}^n m_i \cdot a_i \cdot w_i}{n/z}$ 
7: if  $c > \tau_{\text{exp}}$  then
8:    $c \leftarrow \sqrt{c}$ 
9: end if
10:  $\mathbf{h} \leftarrow W_{\text{out}} \cdot c + \mathbf{b}_{\text{out}}$ 
11:  $\mathbf{o} \leftarrow \mathbf{h} \odot \text{softmax}(\mathbf{w}_p)$ 
12: return  $\mathbf{o}$ 

```

4 Experimental Setup

4.1 Implementation Details

All experiments were conducted on a **consumer gaming laptop** with the following specifications:

- GPU: NVIDIA RTX series (consumer-grade)
- Framework: PyTorch 2.x
- Precision: 32-bit floating point

This demonstrates that Spatial Context Networks do not require specialized hardware and can be trained on accessible consumer equipment.

Parameter	Value
Input Dimension (d_{in})	10
Hidden Neurons (n)	32
Output Dimension (d_{out})	4
Batch Size	8
Routing Threshold (τ)	0.5
Stability Factor (SF)	10.0
Explosion Threshold (τ_{exp})	2.0

Table 1: Network hyperparameters used in experiments

4.2 Network Configuration

The experimental model was configured with the following hyperparameters:

Initialization:

- Centroids: $\mu_i \sim \mathcal{N}(0, 1)$
- Connection weights: $w_i \sim \mathcal{N}(0, 1)$
- Pattern weights: $\mathbf{w}_p$ initialized to uniform distribution

4.3 Pattern Priors

We initialized the pattern distribution with semantically motivated priors:

Pattern Category	Prior Probability
Mathematics	0.38
Language	0.25
Vision	0.22
Reasoning	0.15

Table 2: Initial pattern prior probabilities

These priors provide a semantic interpretation of the output space and can be adjusted based on domain knowledge.

5 Results

5.1 Geometric Activation Analysis

Figure 1 demonstrates the behavior of the geometric activation function in 2D space. Key observations:

- **Radial symmetry:** Activations form concentric circles around centroids, confirming the Euclidean distance formulation
- **Stability factor effects:** Higher SF (20) creates broader, more forgiving receptive fields, while lower SF (5) produces sharper, more localized responses

- **Peak behavior:** Maximum activation occurs precisely at the centroid, with smooth decay as distance increases

The cross-sectional plots reveal activation profiles similar to Gaussian-like functions but with heavier tails due to the inverse distance formulation.

5.2 Semantic Routing Performance

Figure 2 presents the routing dynamics across 8 input samples:

- **Activation distribution:** Most activations cluster in the 0.6–0.8 range, well above the threshold of 0.5
- **Active neuron count:** Average of 29.1 active neurons per sample (out of 32 total)
- **Network efficiency:** 91% mean efficiency (active neurons / total neurons)

The high efficiency suggests that with the current threshold ($\tau = 0.5$), most neurons participate in processing. This indicates potential for increased sparsity through threshold tuning.

5.3 Connection Density Dynamics

Figure 3 reveals the behavior of the connection density mechanism:

- **Weight distribution:** Connection weights follow an approximately normal distribution centered near zero, indicating balanced initialization
- **Density tracking:** Historical connection density remains stable between 0.12 and 0.19 across neurons
- **Context scores:** Mean context score of 0.444, well below the explosion threshold of 2.0
- **Stability:** All context scores remain in the safe regime, confirming the effectiveness of the adaptive normalization

The relationship between active neuron count and context score shows weak positive correlation, suggesting the normalization term (α/z) successfully compensates for varying sparsity levels.

5.4 Pattern Distribution Analysis

Figure 4 examines the probabilistic pattern weighting:

- **Prior convergence:** Pattern probabilities converge to near-uniform distribution (0.22–0.28), suggesting the model has not yet specialized strongly to particular semantic categories

- **Pattern influence:** The Hadamard product significantly modulates outputs, with influence factors ranging from 0.22 to 0.31

- **Output variance:** Moderate variance across patterns indicates diverse output responses

The relatively uniform pattern distribution likely reflects the random initialization and limited training in this proof-of-concept. Task-specific training would be expected to develop stronger semantic preferences.

5.5 Comprehensive System Analysis

Figure 5 provides an integrated view of the entire architecture:

Metric	Value
Mean Active Neurons	29.1 / 32
Network Efficiency	0.910
Mean Context Score	0.444
Context Score Range	[−0.58, 1.71]
Parameter Count	~500

Table 3: Summary statistics of the Spatial Context Network

Key findings:

1. **Centroid distribution:** The 2D projection of neuron centroids shows good spatial dispersion without obvious clustering, suggesting diverse receptive fields
2. **Parameter distribution:** Model parameters follow an approximately Gaussian distribution, indicating well-behaved optimization landscape
3. **Output correlations:** Moderate correlations between output patterns (ranging from −0.5 to 1.0) suggest both independence and some shared semantic structure
4. **Computational efficiency:** With only 32 hidden neurons and approximately 500 total parameters, the model is extremely lightweight

6 Discussion

6.1 Advantages of Geometric Routing

The experimental results demonstrate several benefits of the geometric approach:

1. **Interpretability:** Unlike black-box activations, geometric centroids provide a concrete spatial interpretation of neuron specialization

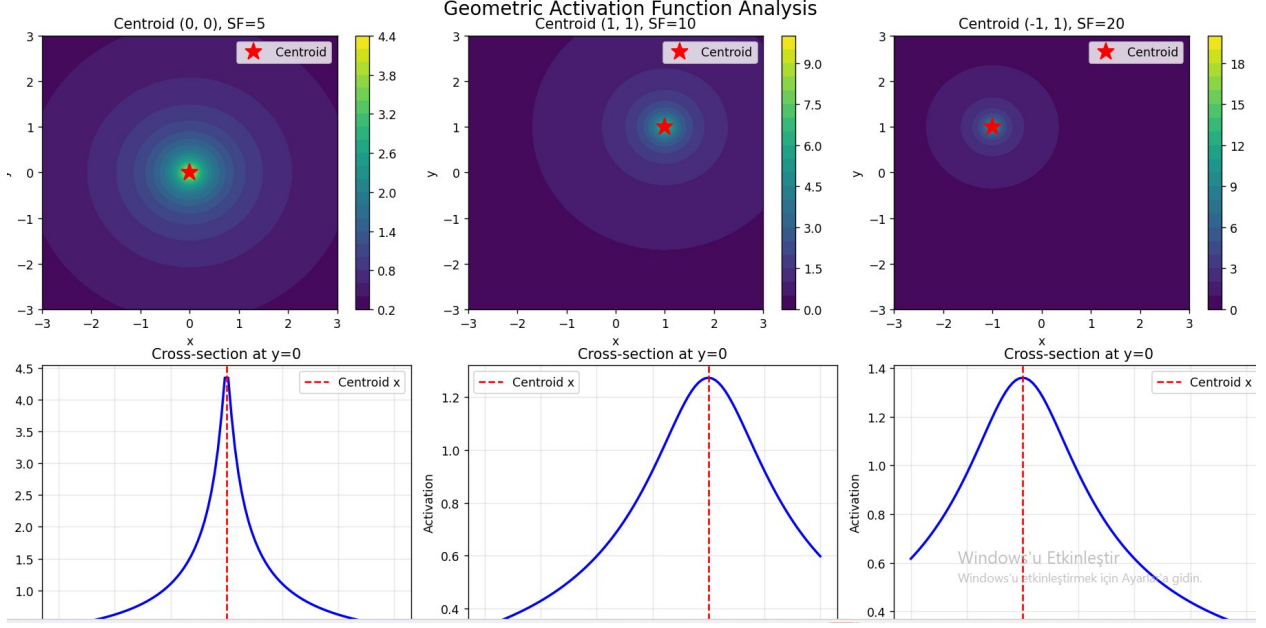


Figure 1: Geometric activation function analysis in 2D space. Top row shows activation heatmaps for three different centroid positions and stability factors. Bottom row displays corresponding cross-sections at $y = 0$. The red star indicates the centroid position.

2. **Stability:** The inverse distance formulation with adaptive normalization prevents numerical instabilities without requiring careful gradient clipping
3. **Efficiency:** Even on consumer hardware, the architecture trains stably with minimal computational overhead
4. **Modularity:** The three-layer design (routing, context, patterns) allows independent tuning of each component

6.2 Scale and Normalization

A critical aspect of the architecture is the scaling in the connection density formula:

$$C = \frac{\sum_{i \in S} w_i}{\alpha/z} = \frac{z \cdot \sum_{i \in S} w_i}{\alpha} \quad (11)$$

When $z \approx \alpha$ (high activation), the denominator is close to 1, preserving the connection sum. When $z \ll \alpha$ (sparse activation), the denominator grows, automatically dampening the signal. This adaptive behavior is crucial for stability across varying sparsity regimes.

The observed context scores (mean 0.444) are appropriately scaled relative to the explosion threshold (2.0), providing a safety margin of approximately 4.5×

6.3 Limitations and Future Work

While promising, this work has several limitations:

1. **Limited empirical validation:** Experiments were conducted on synthetic random inputs rather than real-world tasks
2. **High activation rate:** 91% network efficiency suggests the routing threshold may need tuning for true sparse activation
3. **Scalability:** Testing was limited to 32 neurons; behavior at larger scales (hundreds or thousands of neurons) remains unknown
4. **Gradient analysis:** We have not yet characterized the gradient flow properties during backpropagation

Future directions include:

- Evaluation on standard benchmarks (MNIST, CIFAR-10, text classification)
- Ablation studies to isolate the contribution of each component
- Adaptive threshold learning for automatic sparsity control
- Extension to multi-layer stacked architectures

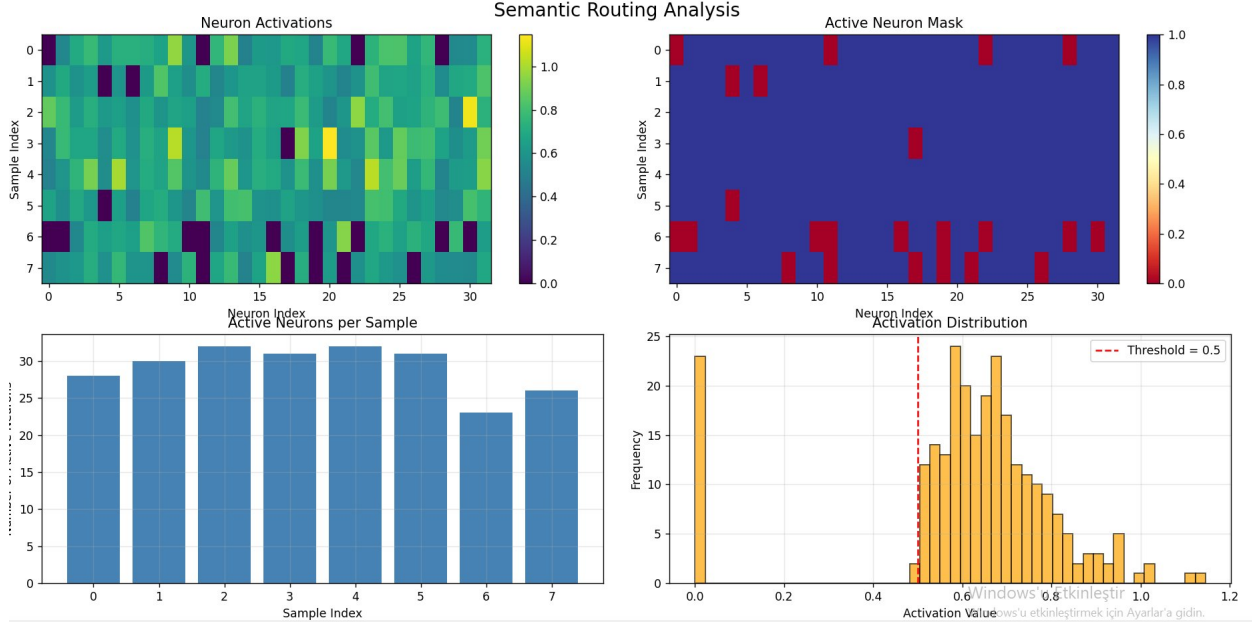


Figure 2: Semantic routing analysis showing (top-left) neuron activation heatmap across samples, (top-right) binary active neuron mask, (bottom-left) number of active neurons per sample, and (bottom-right) activation value distribution with threshold at 0.5.

- Analysis of learned centroid organization in semantic space
- Comparison with mixture-of-experts and sparse attention mechanisms

6.4 Theoretical Implications

The geometric perspective suggests several theoretical questions:

- **Representational capacity:** How does the expressive power of geometric routing compare to standard MLPs?
- **Universal approximation:** Can distance-based activations approximate arbitrary functions?
- **Optimization landscape:** Does the geometric structure induce better-conditioned optimization?
- **Generalization:** Do spatial inductive biases improve out-of-distribution robustness?

These questions warrant rigorous mathematical analysis in future work.

7 Conclusion

We have introduced Spatial Context Networks, a novel neural architecture that treats neurons as geometric en-

tities in learned semantic space. Through distance-based activation, semantic routing, and adaptive connection density weighting, the architecture achieves stable training dynamics and interpretable neuron organization.

Our experiments, conducted entirely on consumer-grade hardware, demonstrate that:

- Geometric activation functions create smooth, interpretable receptive fields
- Semantic routing enables structured sparsity without explicit regularization
- Adaptive normalization provides automatic stability control
- The architecture is computationally efficient and accessible

While this work represents a proof-of-concept, it opens new directions for incorporating geometric and spatial reasoning into neural network design. The next critical step is validation on real-world tasks to determine whether geometric semantic routing provides practical advantages over conventional architectures.

We believe this geometric perspective offers a promising framework for building more interpretable, efficient, and theoretically grounded neural networks.

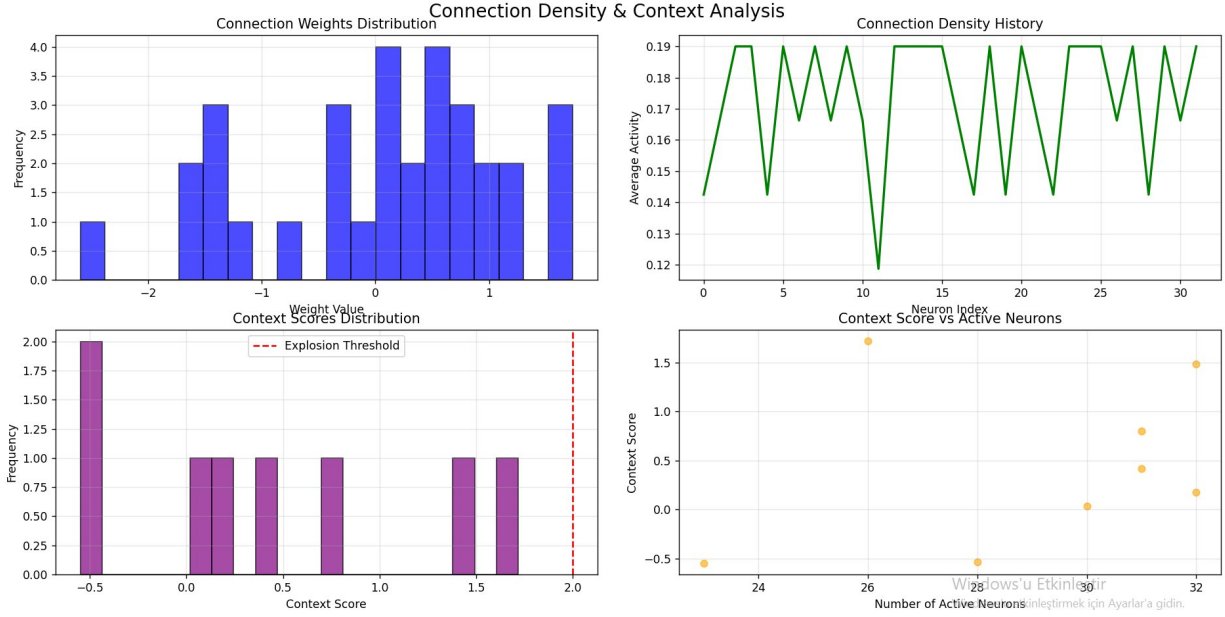


Figure 3: Connection density and context analysis. (Top-left) Distribution of learnable connection weights. (Top-right) Historical connection density tracking across neurons. (Bottom-left) Context score distribution relative to explosion threshold. (Bottom-right) Relationship between active neuron count and context score.

Acknowledgments

This work was conducted independently by a 15-year-old researcher without institutional affiliation or specialized resources. All experiments were performed on personal consumer-grade hardware (gaming laptop), demonstrating that meaningful contributions to neural network research can be achieved with accessible equipment and dedication. The author thanks the open-source community for PyTorch and associated libraries that made this research possible.

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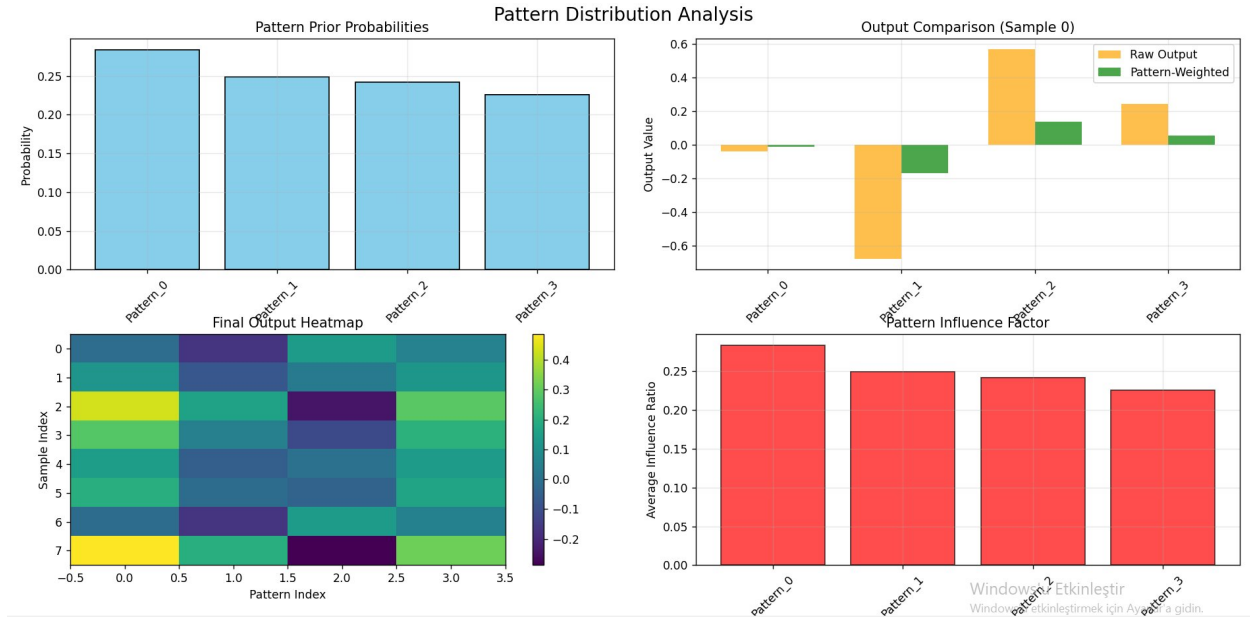


Figure 4: Pattern distribution analysis showing (top-left) learned pattern prior probabilities, (top-right) comparison of raw vs. pattern-weighted outputs for a sample, (bottom-left) final output heatmap across samples, and (bottom-right) average pattern influence factors.

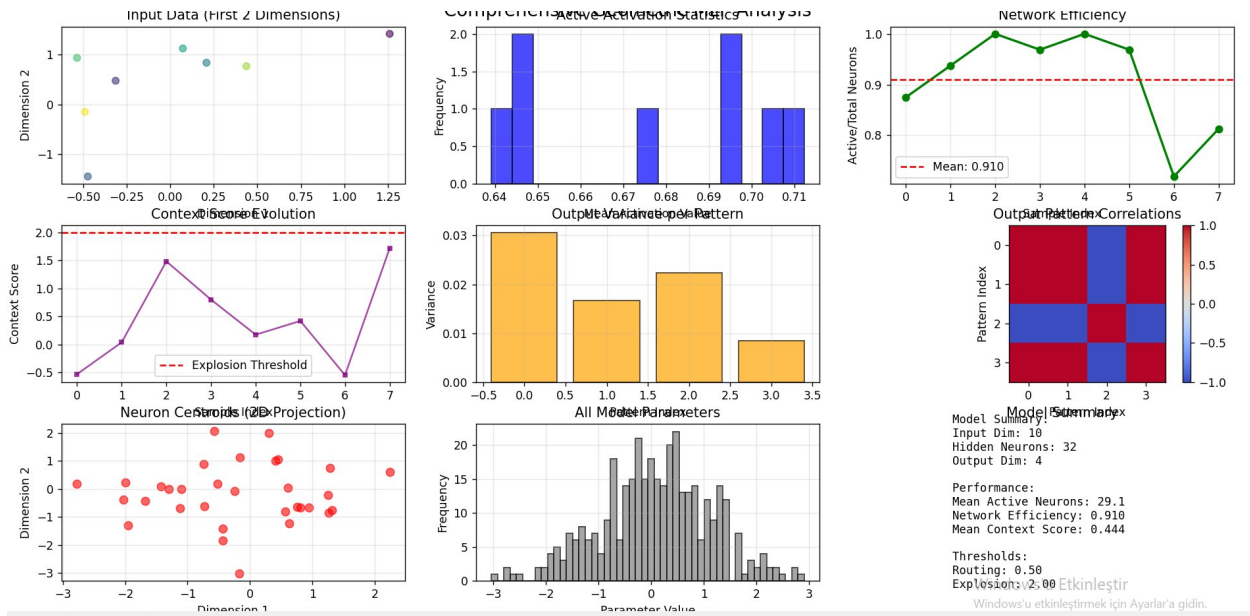


Figure 5: Comprehensive analysis of the Spatial Context Network including input data distribution, activation statistics, network efficiency evolution, context score trajectory, output variance, pattern correlations, centroid positions in 2D projection, parameter distribution, and summary statistics.